

Boss Challenge — Paper 11 (Class 7)

Motion & Time | Comparing Quantities | Electric Current & its Effects | Algebraic Expressions
One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

1. The speed of a moving object is found by dividing the distance it travels by the:
(A) mass of the object
(B) time taken
(C) force applied to it
(D) speed of another object
2. A ratio compares two quantities by:
(A) division
(B) subtraction
(C) addition
(D) multiplication of their squares
3. In an electric circuit, the cell mainly provides the:
(A) light directly
(B) magnetism only
(C) heat only
(D) energy that pushes the current around
4. A symbol such as x , which can stand for different number values, is called a:
(A) equation
(B) constant
(C) coefficient
(D) variable
5. In the SI system, speed is measured using the basic unit called the:
(A) second
(B) metre
(C) metre per second (m/s)
(D) kilometre per hour
6. The ratio 20 : 30 in its simplest form is:
(A) 1 : 2
(B) 20 : 30
(C) 2 : 3
(D) 3 : 2
7. A combination of two or more cells joined together is called a:
(A) fuse
(B) battery
(C) bulb
(D) switch

8. In the expression $3x + 7$, the number 7 is a:
- (A) coefficient of x
 - (B) constant term
 - (C) variable
 - (D) exponent
9. A train moving at 72 km/h has a speed, in metres per second, of:
- (A) 72 m/s
 - (B) 26 m/s
 - (C) 20 m/s
 - (D) 40 m/s
10. 25% of 80 is:
- (A) 20
 - (B) 40
 - (C) 25
 - (D) 16
11. A switch is used in a circuit to:
- (A) make or break the circuit (turn it on or off)
 - (B) measure current
 - (C) store electricity
 - (D) increase the cell's voltage
12. In the term $5y$, the number 5 is called the:
- (A) coefficient of y
 - (B) variable
 - (C) constant
 - (D) power
13. A cyclist covers 30 km in 2 hours. The cyclist's speed is:
- (A) 28 km/h
 - (B) 15 km/h
 - (C) 60 km/h
 - (D) 32 km/h
14. The fraction $\frac{3}{4}$ written as a percentage is:
- (A) 75%
 - (B) 43%
 - (C) 34%
 - (D) 60%
15. In the symbol of a single cell, the longer line stands for the:
- (A) positive terminal
 - (B) switch
 - (C) negative terminal
 - (D) wire

16. Among the following pairs, which one is a pair of like terms?

- (A) $3x$ and $7x$
- (B) x and x^2
- (C) $3x$ and $3y$
- (D) 5 and $5x$

17. A bus moves at a steady 40 km/h . In 3 hours it covers a distance of:

- (A) 80 km
- (B) 43 km
- (C) 120 km
- (D) 13 km

18. The decimal 0.6 written as a percentage is:

- (A) 0.6%
- (B) 600%
- (C) 6%
- (D) 60%

19. Which of these is the best conductor of electricity?

- (A) rubber band
- (B) copper wire
- (C) dry wood
- (D) plastic ruler

20. The sum of $4a$ and $5a$ is:

- (A) $9a^2$
- (B) $20a$
- (C) $9a$
- (D) $45a$

21. A car must cover 150 km at a steady 50 km/h . The time it takes is:

- (A) 5 hours
- (B) 3 hours
- (C) 2 hours
- (D) 100 hours

22. 50% as a fraction in its simplest form is:

- (A) $\frac{1}{5}$
- (B) $\frac{5}{1}$
- (C) $\frac{1}{2}$
- (D) $\frac{1}{50}$

23. Which of these materials is an insulator?

- (A) aluminium
- (B) plastic
- (C) copper
- (D) iron

24. Subtracting $2x$ from $7x$ gives:

- (A) $5x$
- (B) $5x^2$
- (C) $14x$
- (D) $9x$

25. A boy walks 4 km in the first hour and 6 km in the next hour. His average speed for the whole trip is:

- (A) 6 km/h
- (B) 10 km/h
- (C) 2 km/h
- (D) 5 km/h

26. An item bought for ₦200 is sold for ₦250. The profit is:

- (A) ₦250
- (B) ₦450
- (C) ₦200
- (D) ₦50

27. When current flows through a wire and the wire becomes hot, this is called the:

- (A) cooling effect
- (B) chemical effect
- (C) magnetic effect
- (D) heating effect of electric current

28. An expression with only one term, such as $6xy$, is called a:

- (A) trinomial
- (B) binomial
- (C) constant only
- (D) monomial

29. An object is in uniform motion when it covers:

- (A) equal distances in equal intervals of time
- (B) no distance at all
- (C) more distance as time passes
- (D) unequal distances in equal time

30. An item bought for ₦200 is sold for ₦250. The profit percent (on the cost) is:

- (A) 20%
- (B) 25%
- (C) 10%
- (D) 50%

31. The filament of an electric bulb is made of:
- (A) tungsten
 - (B) plastic
 - (C) rubber
 - (D) copper
32. The expression $x + 5$ has two terms, so it is a:
- (A) variable
 - (B) trinomial
 - (C) monomial
 - (D) binomial
33. A car speeding up and slowing down in city traffic is an example of:
- (A) no motion
 - (B) uniform motion
 - (C) circular motion only
 - (D) non-uniform motion
34. A toy bought for ₦120 is sold for ₦90. This is a loss of:
- (A) ₦120
 - (B) ₦210
 - (C) ₦90
 - (D) ₦30
35. The filament of a bulb gives off light because the current:
- (A) makes it magnetic
 - (B) charges it with water
 - (C) cools it down
 - (D) heats it to a very high temperature
36. An expression with three terms, such as $x + y + 7$, is a:
- (A) constant
 - (B) trinomial
 - (C) binomial
 - (D) monomial
37. The time taken by a simple pendulum to complete one full oscillation is called its:
- (A) frequency
 - (B) time period
 - (C) speed
 - (D) amplitude
38. If 5 pens cost ₦60, then 1 pen costs:
- (A) ₦60
 - (B) ₦12
 - (C) ₦5
 - (D) ₦300

39. An electric fuse protects a circuit by:

- (A) increasing the current
- (B) making the bulb brighter
- (C) melting and breaking the circuit when the current is too high
- (D) storing extra current

40. The value of $2x + 3$ when $x = 4$ is:

- (A) 8
- (B) 14
- (C) 11
- (D) 9

41. A simple pendulum makes 20 complete oscillations in 40 seconds, so its time period equals:

- (A) 2 seconds
- (B) 0.5 seconds
- (C) 20 seconds
- (D) 40 seconds

42. If 1 kg of apples costs ₹80, then 3 kg cost:

- (A) ₹83
- (B) ₹160
- (C) ₹80
- (D) ₹240

43. A wire carrying an electric current behaves like a:

- (A) magnet
- (B) fuse
- (C) battery
- (D) insulator

44. The value of $5a - 2$ when $a = 3$ is:

- (A) 15
- (B) 10
- (C) 1
- (D) 13

45. The instrument in a vehicle that shows its speed directly is the:

- (A) odometer
- (B) thermometer
- (C) speedometer
- (D) barometer

46. Simple interest on ₹1000 at 10% per year for 1 year is:

- (A) ₹10
- (B) ₹110
- (C) ₹100
- (D) ₹1000

47. When current flows in a wire placed near a magnetic compass, the compass needle:
- (A) deflects (moves to a new direction)
 - (B) stays exactly still
 - (C) melts
 - (D) catches fire
48. Simplify: $3x + 2x + 4$.
- (A) $6x$
 - (B) $5x + 4$
 - (C) $5x + 4x$
 - (D) $9x$
49. The total distance covered by a vehicle is measured by its:
- (A) speedometer
 - (B) ammeter
 - (C) stopwatch
 - (D) odometer
50. Simple interest on ₦500 at 8% per year for 2 years is:
- (A) ₦40
 - (B) ₦160
 - (C) ₦500
 - (D) ₦80
51. A coil of wire wound on a piece of iron that becomes magnetic only while current flows is an:
- (A) battery
 - (B) electromagnet
 - (C) fuse
 - (D) ordinary bar magnet
52. The expression $3x + 2y$ cannot be simplified further because $3x$ and $2y$ are:
- (A) unlike terms
 - (B) like terms
 - (C) equal
 - (D) constants
53. On a distance-time graph, the motion of an object moving at constant speed appears as a:
- (A) curved line bending up
 - (B) straight slanting line
 - (C) horizontal straight line
 - (D) zig-zag line
54. A plant was 40 cm tall and grew to 50 cm. The percent increase in its height is:
- (A) 10%
 - (B) 50%
 - (C) 20%
 - (D) 25%

55. An electric bell works mainly using the:
- (A) force of gravity
 - (B) magnetic effect of electric current
 - (C) chemical effect
 - (D) heating effect only
56. The perimeter of a square of side s is best written as the expression:
- (A) $s + 4$
 - (B) $2s$
 - (C) s^2
 - (D) $4s$
57. On a distance-time graph, a horizontal (flat) line means the object is:
- (A) moving very fast
 - (B) moving backward fast
 - (C) speeding up
 - (D) at rest (not moving)
58. In a class the ratio of boys to the total number of students is 1 : 4. The percentage of boys is:
- (A) 14%
 - (B) 75%
 - (C) 25%
 - (D) 40%
59. In a simple series circuit with two bulbs, if one bulb fuses, the other bulb will:
- (A) explode
 - (B) glow brighter
 - (C) stop glowing too
 - (D) glow normally
60. Multiplying the term $3x$ by 2 gives:
- (A) $23x$
 - (B) $5x$
 - (C) $3x^2$
 - (D) $6x$
61. On a distance-time graph, a steeper line means the object is moving:
- (A) backward
 - (B) at rest
 - (C) faster
 - (D) slower
62. Which ratio is equivalent to 2 : 5?
- (A) 3 : 5
 - (B) 4 : 10
 - (C) 5 : 2
 - (D) 2 : 10

63. Adding more cells (joined the correct way) to a torch usually makes the bulb glow:
- (A) not at all
 - (B) brighter
 - (C) dimmer
 - (D) the same
64. The value of $x + y$ when $x = 2$ and $y = 5$ is:
- (A) 7
 - (B) 3
 - (C) 25
 - (D) 10
65. Car A goes 100 km in 2 h; Car B goes 100 km in 4 h. The ratio of A's speed to B's speed is:
- (A) 2 : 1
 - (B) 3 : 1
 - (C) 1 : 1
 - (D) 1 : 2
66. Air is about 21% oxygen. In 200 litres of air, the volume of oxygen is about:
- (A) 200 litres
 - (B) 79 litres
 - (C) 21 litres
 - (D) 42 litres
67. Many modern houses use an MCB in place of a fuse. An MCB is a device that:
- (A) stores electricity for the night
 - (B) increases the supply voltage
 - (C) measures the monthly bill
 - (D) switches off automatically when the current is too high
68. In the term $7xy$, the coefficient of xy is:
- (A) 1
 - (B) x
 - (C) 7
 - (D) y
69. A scooter covers 20 km in 30 minutes. Its speed in km/h is:
- (A) 20 km/h
 - (B) 10 km/h
 - (C) 40 km/h
 - (D) 50 km/h
70. A ₹500 shirt is sold at a 10% discount. The discount amount is:
- (A) ₹50
 - (B) ₹100
 - (C) ₹450
 - (D) ₹10

71. A bulb does not glow when the switch is open because the circuit is:
- (A) too cold
 - (B) broken (incomplete)
 - (C) too short
 - (D) complete
72. How many terms are in the expression $2x - 3y + 4$?
- (A) 3
 - (B) 2
 - (C) 4
 - (D) 1
73. A runner's speed is 5 m/s. In km/h this is:
- (A) 18 km/h
 - (B) 5 km/h
 - (C) 1.4 km/h
 - (D) 50 km/h
74. A ₹500 shirt is sold at a 10% discount. Its selling price is:
- (A) ₹450
 - (B) ₹550
 - (C) ₹50
 - (D) ₹490
75. Which appliance mainly uses the heating effect of electric current?
- (A) an electric room heater
 - (B) a radio antenna
 - (C) a compass
 - (D) an electric bell
76. A car travels at a steady speed v for a time t . The distance it covers is written as the expression:
- (A) $v - t$
 - (B) v / t
 - (C) $v \times t$
 - (D) $v + t$
77. Which of these usually moves the fastest?
- (A) a bullock cart
 - (B) a person walking
 - (C) a bicycle
 - (D) an aeroplane
78. What percent is 15 out of 60?
- (A) 45%
 - (B) 15%
 - (C) 60%
 - (D) 25%

79. To keep ourselves safe, we should never touch electrical switches with:

- (A) dry hands
- (B) a plastic glove
- (C) wet hands
- (D) a dry cloth

80. If one pen costs p , the cost of 5 such pens is written as:

- (A) $p + 5$
- (B) $p - 5$
- (C) $5p$
- (D) $p/5$

81. If a simple pendulum is made longer, its time period generally:

- (A) becomes zero
- (B) increases
- (C) decreases
- (D) stays exactly the same

82. 100 is shared between two friends in the ratio 1 : 4. The smaller share is:

- (A) 40
- (B) 80
- (C) 20
- (D) 25

83. The strength of an electromagnet can be increased by:

- (A) using fewer cells
- (B) removing the iron core
- (C) increasing the number of turns in the coil
- (D) cutting the wire

84. Simplify: $8y - 3y + y$.

- (A) $6y^2$
- (B) $4y$
- (C) $6y$
- (D) $12y$

85. A train travels 120 km in 2 h, then 60 km in 1 h. Its average speed for the journey is:

- (A) 90 km/h
- (B) 60 km/h
- (C) 45 km/h
- (D) 180 km/h

86. Car A's speed is 60 km/h and Car B's is 75 km/h. Car B is faster than Car A by:

- (A) 25%
- (B) 75%
- (C) 15%
- (D) 20%

87. A torch uses 2 cells of 1.5 V each, joined in a line. The total voltage of the battery is:

- (A) 2 V
- (B) 1.5 V
- (C) 3 V
- (D) 0.75 V

88. The value of $4x + 9$ when $x = 0$ is:

- (A) 13
- (B) 9
- (C) 0
- (D) 4

89. Two towns are 240 km apart. A car at a steady 60 km/h takes how long to drive between them?

- (A) 3 hours
- (B) 4 hours
- (C) 6 hours
- (D) 2 hours

90. The fraction $\frac{5}{8}$ written as a percentage is:

- (A) 58%
- (B) 62.5%
- (C) 40%
- (D) 85%

91. If each cell gives 1.5 V, the total voltage of 4 such cells joined in a line is:

- (A) 6 V
- (B) 4 V
- (C) 1.5 V
- (D) 5.5 V

92. Simplify: $2a + 3b + 4a$.

- (A) $6a + 3b$
- (B) $9ab$
- (C) $5a + 4b$
- (D) $6ab$

93. An object moving at a steady 10 m/s for 5 s covers a distance of:

- (A) 2 m
- (B) 10 m
- (C) 50 m
- (D) 15 m

94. ■2000 is kept at 5% per year simple interest for 1 year. The total amount after 1 year is:

- (A) ■100
- (B) ■2100
- (C) ■2000
- (D) ■2500

95. Conventional current in a circuit is taken to flow, outside the cell, from the cell's:
- (A) switch to the fuse only
 - (B) negative to positive outside the cell
 - (C) positive terminal to its negative terminal
 - (D) bulb to the wire only
96. If each cell of a battery gives a volts, the total voltage of n such cells joined in a line is written as:
- (A) a / n
 - (B) $n + a$
 - (C) $a - n$
 - (D) $n \times a$
97. To measure short time intervals in races we most commonly use a:
- (A) stopwatch
 - (B) calendar
 - (C) measuring tape
 - (D) weighing scale
98. 10% of 10% of 1000 is:
- (A) 1
 - (B) 200
 - (C) 10
 - (D) 100
99. To measure the electric current flowing in a circuit, we connect an:
- (A) speedometer
 - (B) voltmeter
 - (C) thermometer
 - (D) ammeter
100. The area of a rectangle of length l and breadth b is written as the expression:
- (A) $2(l + b)$
 - (B) $l \times b$
 - (C) $l - b$
 - (D) $l + b$